Cloud-Based Smart Warehouse Automation

Simulation-Based Intelligent Warehouse Automation System

CSA1501 cloud computing

Guide By:

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ABSTRACT

- A cloud-deployable warehouse automation platform combining a FastAPI/PostgreSQL backend with a full multi-robot fleet simulation and a 2D Digital Twin.
- Real-time A* path planning with obstacle avoidance, cell-reservation collision detection, and priority-based conflict resolution drives autonomous robot navigation.
- An explainable AI layer forecasts demand (WAPE/MAE/RMSE/sMAPE with walk-forward validation) and flags inventory shrinkage using Isolation Forest anomaly detection.
- A Scenario Lab lets managers run reproducible, seeded what-if experiments — including OR-Tools optimized scheduling and SimPy discrete-event packing simulations.
- Security is enforced through JWT/RBAC authentication, step-up OTP verification, and a hash-chained, tamper-evident audit ledger.
- System health, metrics, and error tracking are exposed through Prometheus, Grafana, and Sentry for full observability.

INTRODUCTION

- Manual and semi-automated warehouses struggle with robot congestion, unpredictable demand, and undetected inventory shrinkage.
- This project builds a single integrated platform where a simulated robot fleet, inventory intelligence, and warehouse security work together end to end.
- A tick-based Digital Twin keeps a live, database-reconciled view of every robot, task, and obstacle in the warehouse.
- Machine learning adds forecasting and anomaly detection on top of raw stock-movement data, so decisions are explainable, not black-box.
- The system is built to be demoable and auditable: every simulation run is seeded and reproducible, every security-relevant action is logged to an immutable ledger.

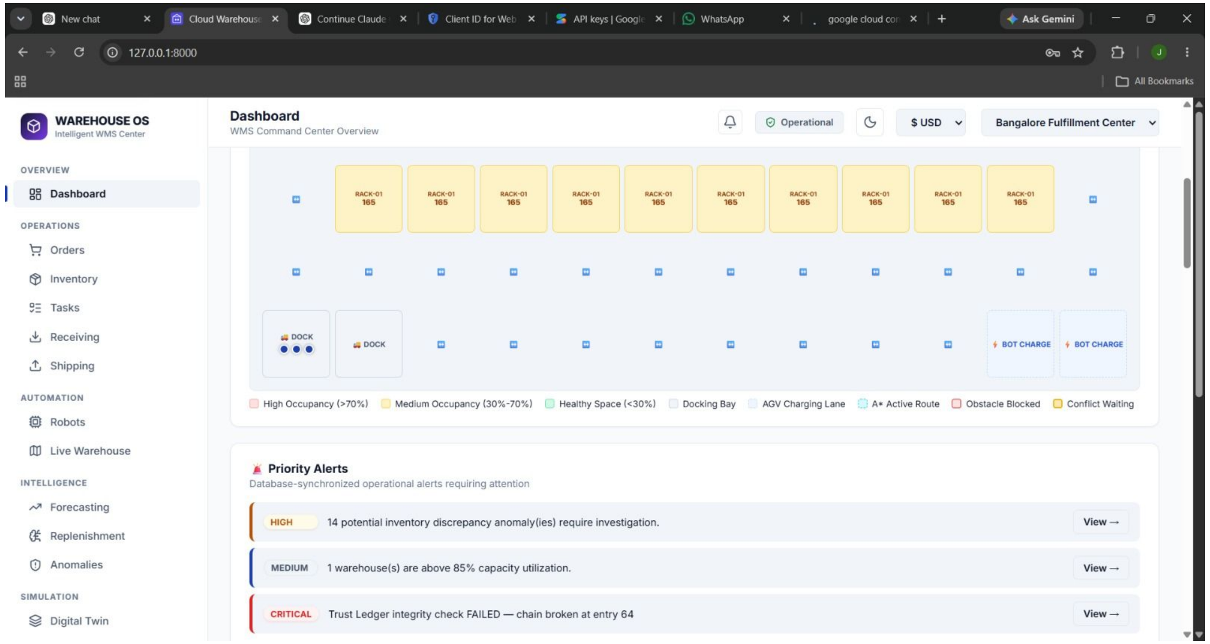
OBJECTIVES

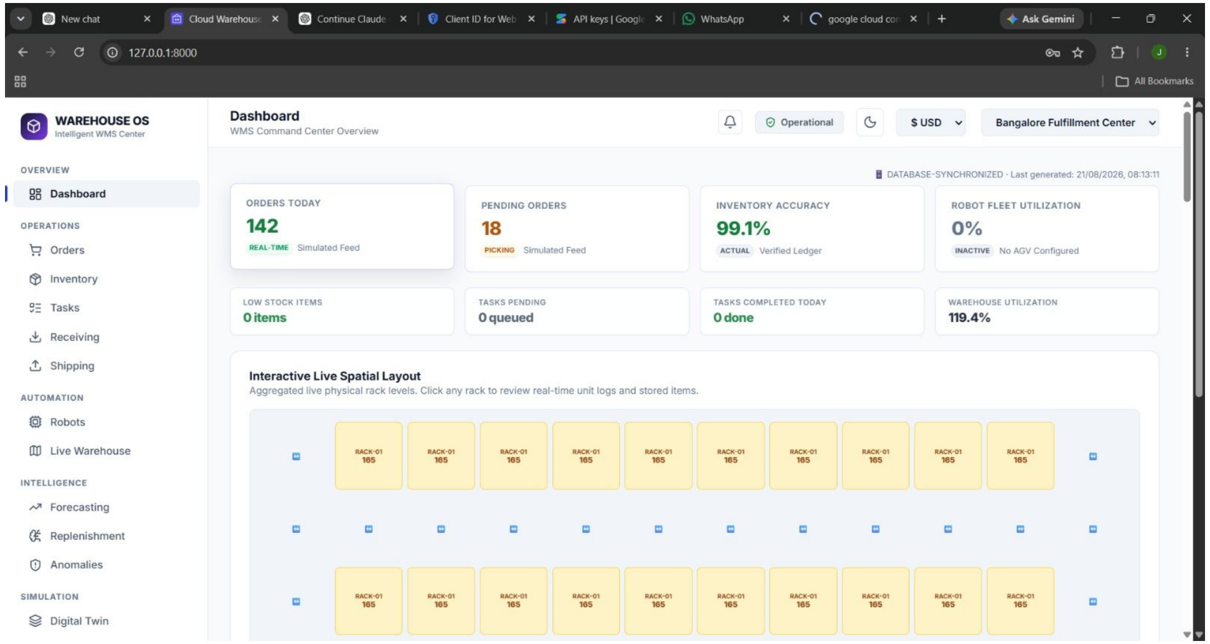
- To simulate a multi-robot warehouse fleet with A* path planning, dynamic replanning, and collision-free navigation.
- To build a 2D Digital Twin that visualizes robot state, routes, obstacles, and warehouse KPIs in real time.
- To forecast product demand and detect inventory anomalies using explainable machine learning models.
- To secure the platform end to end with RBAC, step-up authentication, and a hash-chained audit ledger, while exposing full observability through metrics and error tracking.

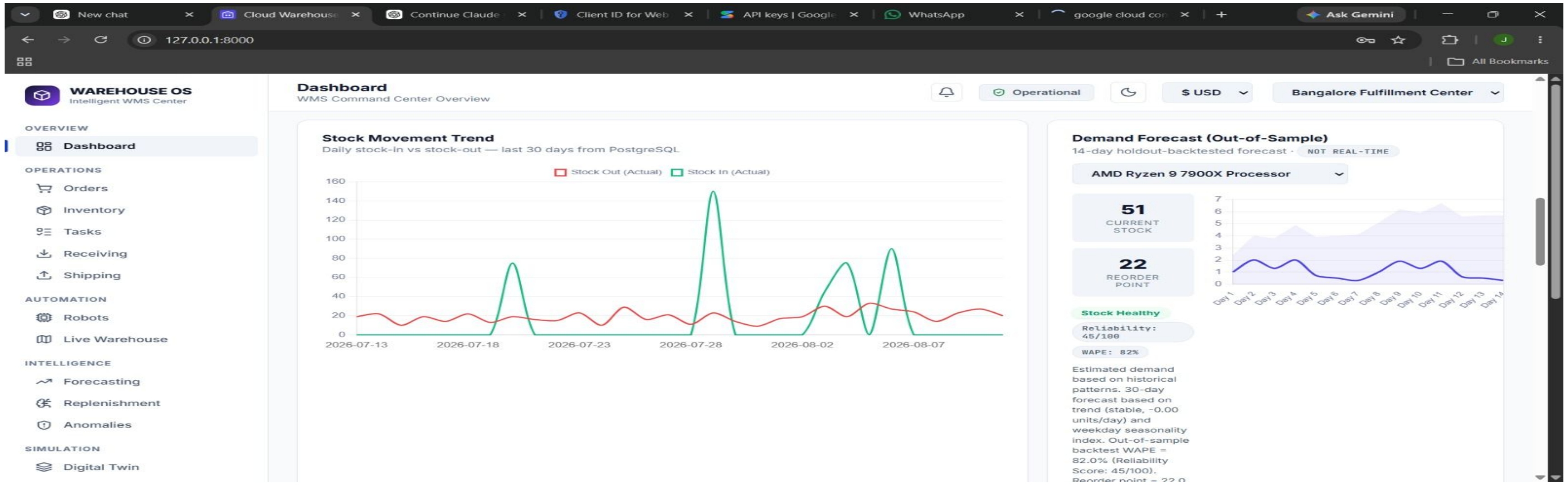
Module 1: Multi-Robot Fleet Simulation & Path Planning

- Implements A* pathfinding with a Manhattan-distance heuristic over a weighted, obstacle-aware warehouse grid.
- Detects and resolves same-cell and head-on collisions between robots using a tick-indexed reservation table.
- Applies priority-based conflict resolution so lower-priority robots yield and wait rather than collide.
- Triggers dynamic replanning when a robot is blocked for multiple ticks, generating a fresh route around the obstruction.
- Assigns tasks to robots using cost-based, battery-aware scoring so low-charge robots are not routed onto long jobs.

Module 1 : Output



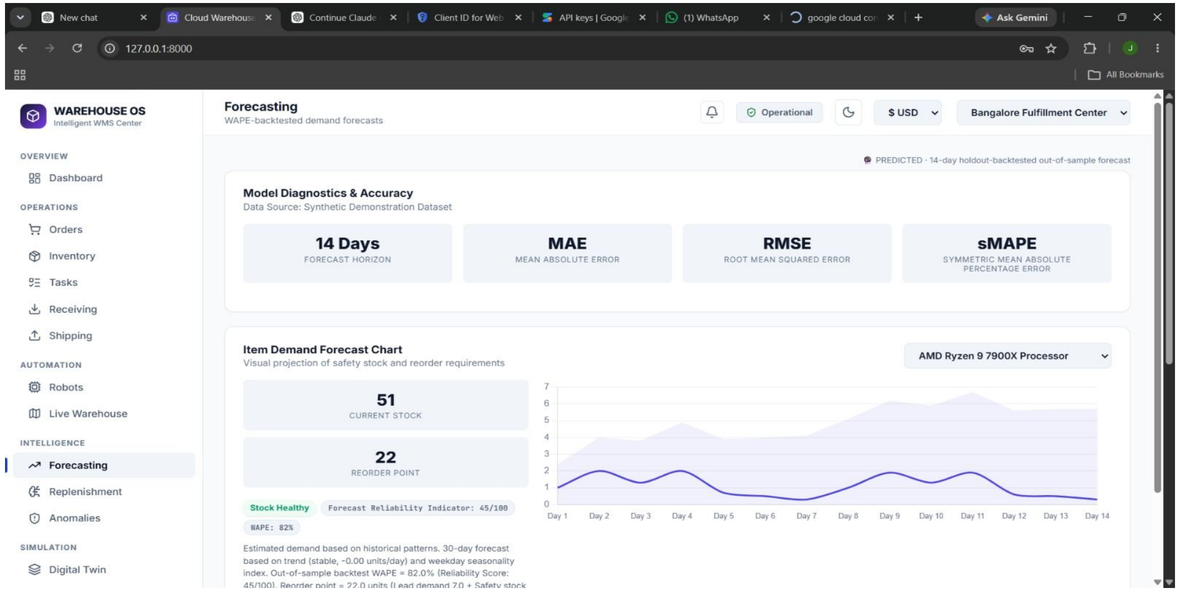


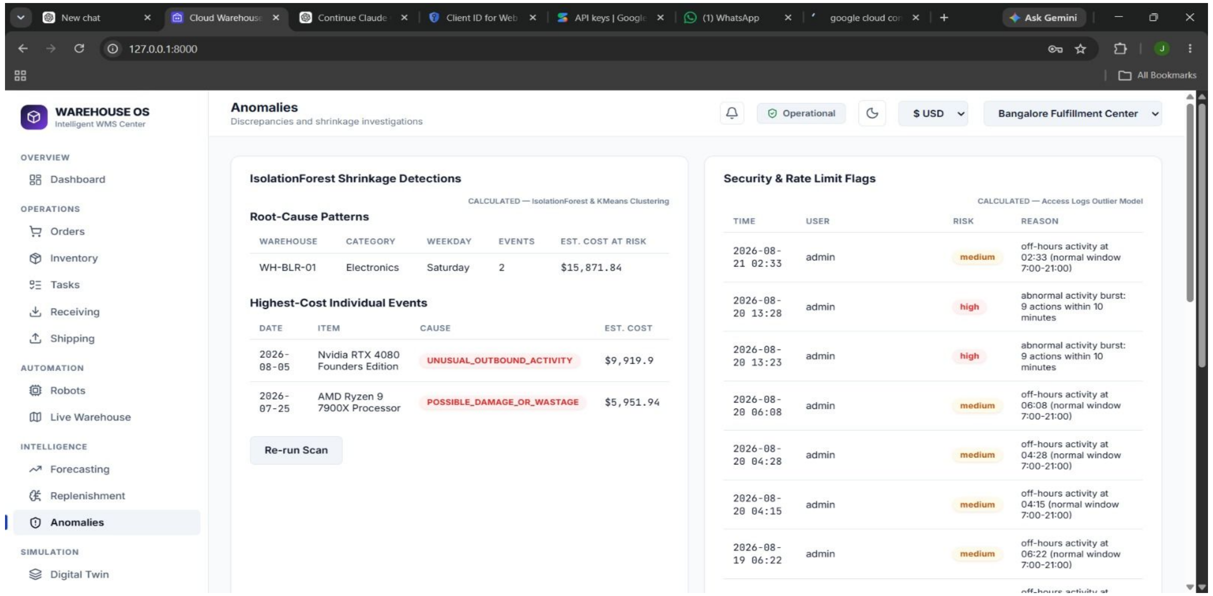


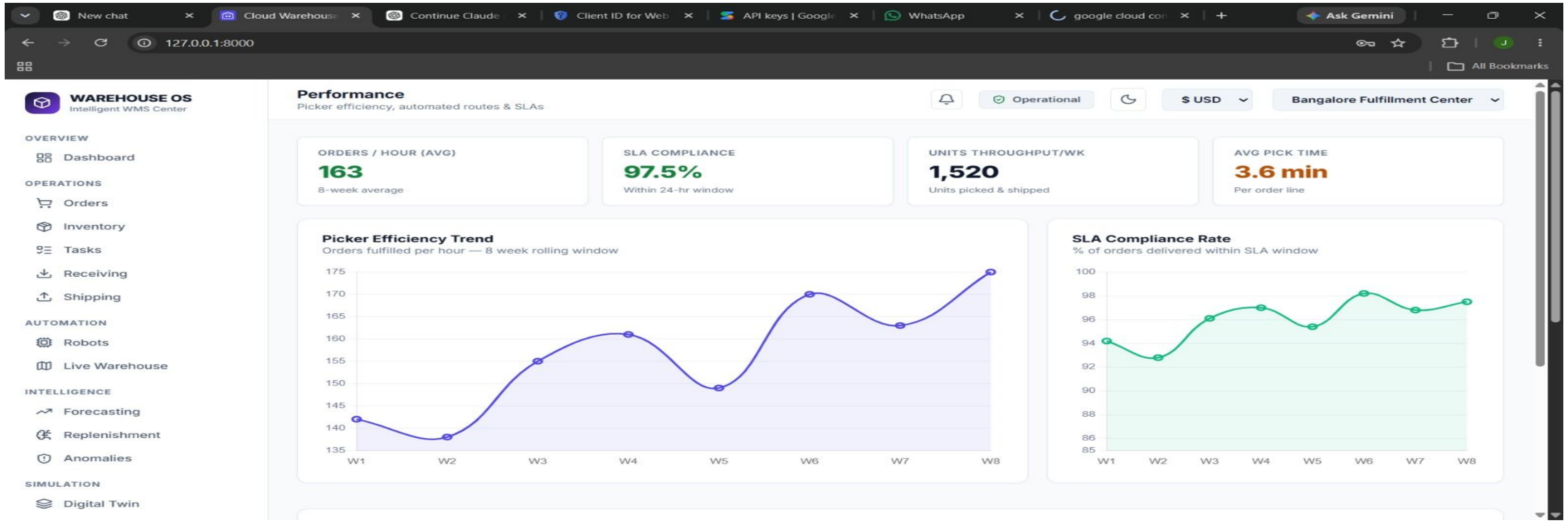
Module 2: Demand Forecasting & Explainable Anomaly Detection

- Forecasts short-horizon product demand using chronological train/test splits and walk-forward validation against naive and moving-average baselines.
- Scores forecast quality with WAPE, MAE, RMSE, and sMAPE, and attaches a composite reliability score to every forecast.
- Runs an Isolation Forest model over stock-movement data to flag inventory anomalies that may indicate shrinkage, theft, or damage.
- Performs ABC inventory classification to prioritize which items need the closest monitoring and fastest replenishment.
- Surfaces every AI recommendation with a plain-language explanation and a human approval step before it is acted on.

Module 2: Output







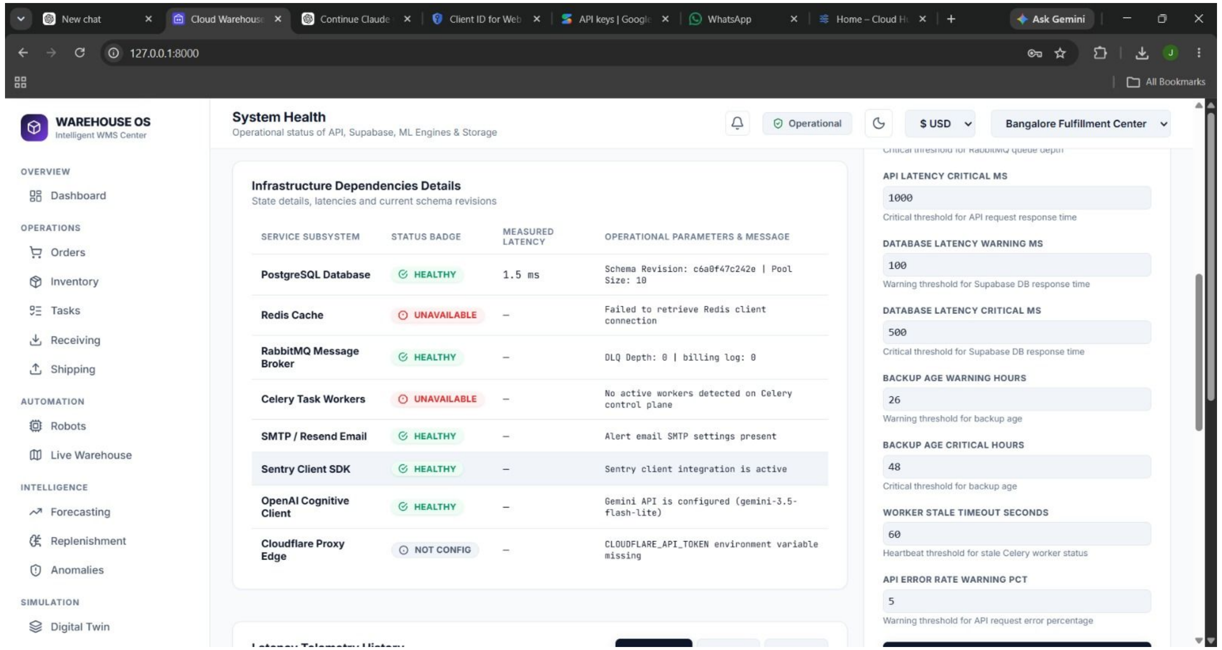
Module 3: 2D Digital Twin & Scenario Lab

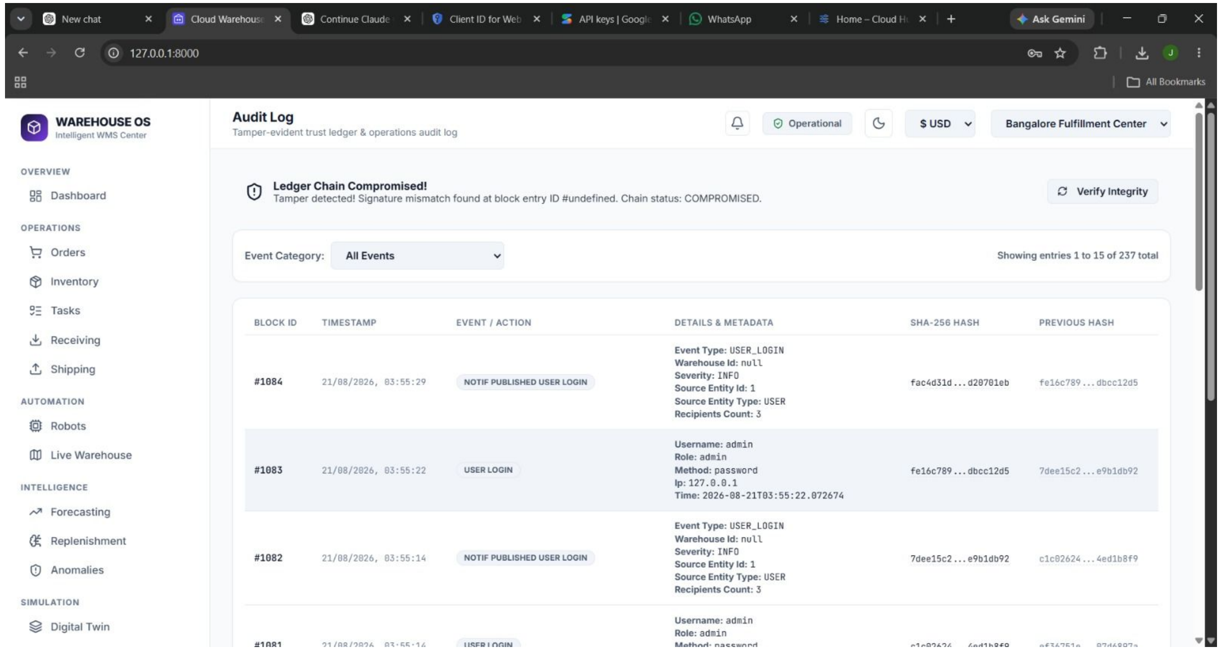
- Maintains a database-reconciled 2D Digital Twin showing live robot positions, routes, obstacles, and warehouse KPIs.
- Provides Start / Pause / Step / Resume / Reset simulation controls, all wired to real backend endpoints.
- Runs reproducible, seed-based What-If experiments through a dedicated Scenario Lab, with full experiment history and result export.
- Benchmarks robot-task assignment using Google OR-Tools (CP-SAT) against the warehouse's real, live robot fleet.
- Models packing-station throughput and operator contention using a discrete-event SimPy simulation.

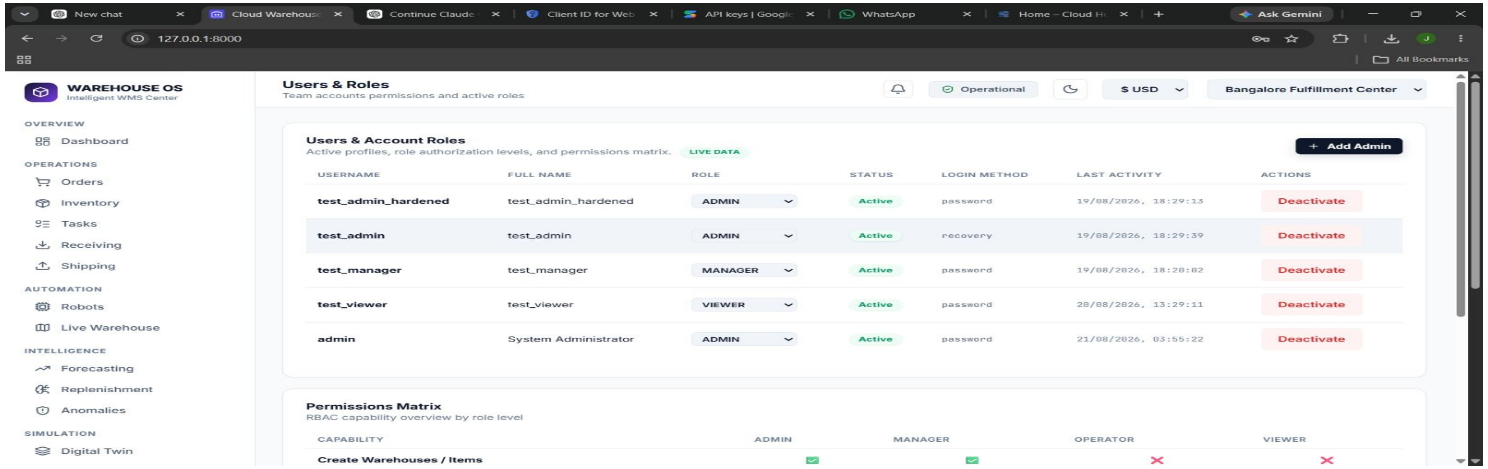
Module 4: Security, Audit & Observability

- Enforces Role-Based Access Control with granular permissions, bcrypt password hashing, and JWT-based sessions.
- Requires step-up OTP verification for sensitive actions, with account lockout after repeated failed login attempts.
- Logs every security- and inventory-relevant event to a hash-chained, tamper-evident audit ledger with built-in integrity verification.
- Exposes real-time application metrics via Prometheus and visualizes them on Grafana dashboards.
- Tracks application errors and performance through Sentry, with a system health endpoint reporting live status of every connected service.

Module 4: Output







PROJECT COMPONENT REQUIREMENTS

Hardware Requirements

- Intel Core i5 Processor or higher
- 8 GB RAM
- 256 GB SSD
- Laptop/Desktop with Internet Connection
- Cloud Database (PostgreSQL, Render-hosted)

Software Requirements

- Windows 10/11
- Python 3.11+
- Visual Studio Code
- FastAPI Framework
- PostgreSQL / SQLite (testing only)

Core Libraries

- SQLAlchemy, Alembic
- scikit-learn, pandas, NumPy
- Google OR-Tools, SimPy

Dataset Requirements

- Synthetic Warehouse Inventory Dataset
- Multi-Warehouse Stock-Movement Logs
- Simulated Robot Telemetry & Task Events

TASKS

- Design the warehouse grid, robot fleet model, and A* path-planning engine with collision detection.
- Build the FastAPI backend, PostgreSQL schema, and Alembic migrations for orders, tasks, robots, and inventory.
- Train and validate the demand-forecasting and shrinkage-detection models, and expose them through backend endpoints.
- Build the 2D Digital Twin, Scenario Lab, and AI Decision Center frontend, wiring every module end to end.
- Implement RBAC, OTP, audit ledger, and observability (Prometheus, Grafana, Sentry), and validate with automated tests.

SPECIFIC PARAMETERS

- **Robot State:** Position, battery level, task assignment, and status (idle / moving / waiting / failed), tracked per tick.
- **Path Cost:** Weighted grid cost combined with the Manhattan heuristic to compute each robot's route.
- **Product Demand:** Historical stock-movement volume used as the basis for the forecast model.
- **Inventory Deviation:** Difference between expected and actual stock levels, used to flag anomaly events.
- **Task Priority:** Composite score combining order urgency, robot cost, and battery level for assignment.

OUTCOME PARAMETERS

- **Route & Collision Report:** Shows generated robot routes and any conflicts resolved through waiting or replanning.
- **Forecast Reliability Score:** Displays demand forecast accuracy against naive and moving-average baselines.
- **Shrinkage Risk Classification:** Classifies flagged anomalies as likely shrinkage, damage, or data error.
- **Scenario Comparison Report:** Compares baseline vs. OR-Tools optimized scheduling across repeated simulation runs.
- **Audit & System Health Digest:** Summarizes ledger integrity status and live connection status of every integrated service.

TECHNIQUES TO BE USED

- **Robotics & Path Planning:** A* search, Manhattan heuristic, cell-reservation collision avoidance.
- **Machine Learning:** Isolation Forest for anomaly detection; walk-forward validated demand forecasting.
- **Optimization & Simulation:** Google OR-Tools (CP-SAT) scheduling and SimPy discrete-event simulation.
- **Backend & Data:** FastAPI for the API layer, SQLAlchemy/PostgreSQL for persistence, Alembic for migrations.
- **Security & Observability:** RBAC, OTP step-up auth, hash-chained audit ledger, Prometheus, Grafana, Sentry.

FUTURE SCOPE

- Integrate real IoT/RFID sensors for live robot telemetry instead of simulated data.
- Deploy the async task pipeline on live Redis and RabbitMQ infrastructure for true background processing at scale.
- Improve forecast accuracy using deep-learning models trained on larger historical datasets.
- Extend the Digital Twin from 2D to a full 3D warehouse visualization.
- Add a native mobile app for warehouse managers with push notifications for critical alerts.

CONCLUSION

- The platform simulates a full multi-robot warehouse fleet with real-time, collision-free path planning.
- It forecasts demand and detects shrinkage using explainable, validated machine learning rather than black-box alerts.
- The Scenario Lab enables reproducible, seed-based experimentation and optimization comparison.
- Security and observability are built in from the ground up, not bolted on: RBAC, audit ledger, metrics, and error tracking all work end to end.
- Overall, the project delivers a practical, auditable, and reproducible simulation platform for intelligent warehouse automation.

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Thank You